

SUPPORT SIGNALS

WHAT NORMAL LOOKS LIKE

Supports are your landmarks and a common place for damage. Know what normal looks like so the odd one stands out.

KEY IDEA

Every support of the same type should look nearly identical. The odd one out is the one to look at.

1 WHAT THEY ARE

- Tube support plates, baffles, and anti-vibration bars (AVBs) in the U-bend.
- Usually carbon steel, which is magnetic, so they make a large signal.
- Stainless steel supports make a much smaller signal.

2 HOW THEY SHOULD LOOK

- A big signal, usually much larger than a typical flaw.
- Strongest on the low frequencies.
- Shape depends on plate type (drilled or broached), material, and frequency.

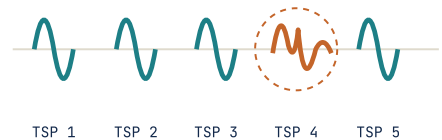
3 HOW THEY SHOULD ACT

- Regular spacing that matches the drawing. Count them to know where you are.
- Same shape, size, and angle at every support of the same type.
- After a good mix, only a small, similar residual remains at each one.

4 WHAT TO LOOK FOR

- Distortion: an extra loop, a lopsided shape, or a tail the neighbors don't have.
- A larger mix residual than the other supports.
- Wear from rubbing, best seen on the mix, within the support width.
- Dents (DNT), and deposits or sludge, especially on the lower plates.
- A missing or extra support. Re-check your count.

odd one out



Supports in a tube should all match. The distorted one gets a closer look (illustrative).

COMPARE AGAINST

- › Neighboring supports in the same tube
- › The same support in nearby tubes
- › The previous inspection, if you have it

QUICK TIPS

Count supports from a known landmark in every tube.

Switch to the mix before calling anything at a support.

A bigger residual than its neighbors deserves a second look.